

To: Consumer Management Specialist

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Topic: Predicting Poisonous and Edible Mushrooms

Goal:

The purpose of this report is to introduce the best method for determining if mushrooms are poisonous or edible based on their characteristics. An automated system was developed to predict whether specific mushrooms are poisonous to reduce risk to consumers. This system analyzes key characteristics of mushrooms as independent variables to classify them as edible or poisonous. This system is a high accuracy model that is best used for classification problems when reliable identification is required.

Methods used:

The methods I used for this analysis were a Decision Tree model and Random Forest model to solve the classification problem on whether a mushroom is poisonous or edible. To evaluate the models, the dataset was split into testing and training datasets. The training dataset was used to build the models, while the testing dataset was used to evaluate how the models performed on new data. Each model's performance was measured based on confusion matrix and accuracy rates, which identified correct classification as well as errors.

The Random Forest model improves the Decision Tree model as it creates multiple decision trees from sample training datasets. This process creates model stability as it reduces overfitting, allowing for better model analysis.

Data collection:

An analysis was conducted using a dataset of approximately 60,000 samples of mushroom characteristics to solve the binary classification problem on whether mushrooms are poisonous or edible. The dataset includes both numerical and categorical variables with characteristics of mushrooms including cap, gill and stem features. By using these variables, we can create a predictive model that allows users to classify mushrooms.

All categorical variables were converted to dummy variables to ensure proper analysis within the model. Missing variables were included in the dataset which were handled as a separate category to prevent loss in data. The target variable of "class" determines whether a mushroom is poisonous or edible based on its independent variables.

Results:

The results of each model indicate the accuracy of predicting the classification of mushrooms. The Decision Tree model resulted in approximately 99.86% accuracy, while the Random Forest model reached 100% accuracy. As this is a high-risk assessment, the model with the highest accuracy rate with no false negatives is essential. It was determined that the Random Forest model was best fit for this analysis, as it automates predictive outcomes with high accuracy.

When running our Decision Tree Model, it was determined that there were 4 false negatives. This means that 4 mushrooms were classified as edible but were poisonous. The error from this model is high risk, as it can lead to dangerous consumer outcomes. The Decision Tree model produced 22 false positives as well, meaning 22 edible mushrooms were classified as poisonous.

Our Random Forest model classified all mushrooms correctly with 100% accuracy, producing zero false positives and zero false negatives. Our Random Forest model is more reliable for predictions which minimizes the risk of misclassification in new datasets.

Model Results	Column1	Column2
Model	Accuracy	False Negatives
Decision Tree	0.998581	4
Random Forest	1.00	0

Figure 1: Model Results

Confusion Matrix (Decision Tree)	Column1	Column2
	Predicted Poisonous	Predicted Edible
Actual Poisonous	8150	4
Actual Edible	22	10145

Figure 2: Decision Tree Confusion Matrix

Confusion Matrix (Random Forest)	Column1	Column2
	Predicted Poisonous	Predicted Edible
Actual Poisonous	8154	0
Actual Edible	0	10167

Figure 3: Random Forest Confusion Matrix

Final Model Choice:

The best model choice is the Random Forest model. This model achieved 100% accuracy and can reliably determine whether mushrooms are poisonous or edible. By combining multiple decision trees, the Random Forest model offers predictions that reduce critical errors compared to the single Decision Tree model.

The Random Forest model resulted in zero false negatives in comparison to the Decision Tree model which resulted in 4 false negatives. This is important as a single Decision Tree could create classification

errors when new data is introduced, leading to consumer risk. The low error rate of the Random Forest model is more likely to maintain accuracy with unseen data. Since the Random Forest model classifies mushroom edibility with 100% accuracy, this will be the most appropriate choice.